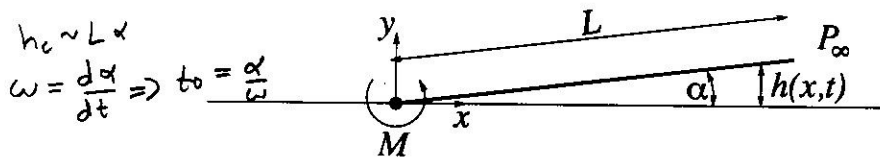


INTRODUCTION TO FLUID MECHANICS

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✓ F13 A plate of length L is initially sitting on a horizontal plane in the presence of a stagnant liquid atmosphere with reduced pressure P_∞ . At a given instant, we begin to rotate the plate with constant angular velocity $\omega = d\alpha/dt$ by applying a given torque M at its left end, as sketched in the figure. For the analysis, use the approximation $h(x, t) = x \tan(\alpha) \simeq x\alpha$, valid for $\alpha \ll 1$.

1. Demonstrate that for values of α sufficiently smaller than a critical value, to be determined, the fluid motion in the gap formed between the plate and the wall is dominated by viscosity.
2. Obtain the velocity profile v_x in the gap as a function of the unknown value of $P_l(x, t) = -\partial P/\partial x$ as well as the associated volume flux at a given section $Q = \int_0^{\alpha x} v_x dy$.
3. Using continuity, write an equation linking Q and ω , and integrate it to compute the pressure distribution $P(x, t)$ (in the integration, you may anticipate that $x^3 P_l \rightarrow 0$ as $x \rightarrow 0$).
4. Determine the torque $M(t)$ needed to provide a constant angular velocity ω .



$$h_c \sim L\alpha$$

$$\omega = \frac{d\alpha}{dt} \Rightarrow t_0 = \frac{\alpha}{\omega}$$

$$1) \frac{h_c^2 / (\mu \omega)}{t_0} \sim \frac{L^2 \alpha^2 \omega}{\mu \alpha} \ll 1 \Rightarrow \alpha \ll \frac{\nu}{L^2 \omega}$$

$$\frac{v_L h_c}{\nu} \frac{h_c}{L} \sim \frac{\omega L^2}{\nu} \frac{h_c}{L} \sim \frac{\omega L^2 \alpha}{\nu} \ll 1 \Rightarrow$$

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0 \Rightarrow v_L \sim \frac{\omega L^2}{h_c}$$

$$2) v_x = \frac{P_l}{2\mu} y(h-y) \rightarrow Q = \frac{P_l}{12\mu} h^3 = -\frac{x^3 \alpha^3}{12\mu} \frac{\partial P}{\partial x}$$

$$3) \int_0^h \frac{\partial v_x}{\partial x} dy + \int_0^h \frac{\partial v_y}{\partial y} dx = 0 \Rightarrow \frac{\partial}{\partial x} Q - \frac{dh}{dx} v_x(y=h) + v_y(h) - v_y(0) = 0$$

$$\frac{\partial Q}{\partial x} + \omega x = 0$$

$$\frac{\partial}{\partial x} \left(\frac{\alpha^3}{12\mu} x^3 \frac{\partial P}{\partial x} \right) - \omega x = 0 \Rightarrow \frac{\alpha^3}{12\mu} x^3 \frac{\partial P}{\partial x} - \frac{\omega x^2}{2} = 0$$

$$\frac{\partial P}{\partial x} = \frac{6\mu\omega}{\alpha^3} \frac{1}{x} \Rightarrow \boxed{P - P_\infty = \frac{6\mu\omega}{\alpha^3} \ln\left(\frac{x}{L}\right)}$$

$$4) \bar{M}_{F \rightarrow \text{plate}} = - \int_{\Sigma_F} (P - P_\infty) \bar{x} \wedge \bar{n} d\sigma = \int_0^L \frac{6\mu\omega}{\alpha^3} x \ln \frac{x}{L} dx \bar{e}_z = -\frac{3}{2} \frac{\mu\omega L^2}{\alpha^3} \bar{e}_z$$

TO COUNTERBALANCE THIS MOMENTUM WE NEED TO APPLY A TORQUE $\bar{M} = \frac{3}{2} \frac{\mu\omega L^2}{\alpha^3} \bar{e}_z$